Phase 0 Architecture Audit: Mechanistic Discovery Framework

System Decomposition, Parameter Inventory, and Operational Boundaries

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November 6, 2026

Research Mandate: Reverse-engineer language-grounded world models to extract causally essential circuits for FPGA acceleration.

Parameter Inventory: 100% Accountability Across Architectures

One Core Insight

Every trainable tensor (72 parameters across 3 models) has been mapped, classified, and verified against SHA-256 checkpoint fingerprints.

- **Baseline Navigation Policy**: 147,028 parameters ($z \in \mathbb{R}^{32}$, 1-step direct visual policy).
- **LeWorldModel (LeWM)**: 148,913 parameters (Conv4-LN encoder + action-conditioned residual transition core + SIGReg).
- **H-JEPA**: 166,037 parameters (Shared visual encoder + Macro Predictor at stride $K = 5$ + Micro tracking policy).

Model	Parameters	Status
Baseline Policy	147,028	Verified
LeWorldModel	148,913	Verified
H-JEPA	166,037	Verified
Total Audited	**461,978**	**100% Accounted**

Integrity Check

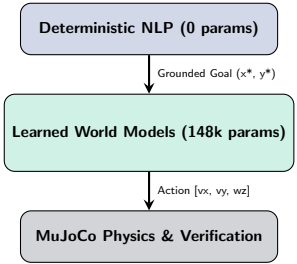
All checkpoints SHA-256 hashed and locked in `research/audit/parameter_inventory.csv`.

System Boundary: Separating Learned Intelligence from Heuristics

One Core Insight

Language parsing and flank routing are deterministic software, whereas visual dynamics and collision prediction are learned neural computations.

- **Deterministic NLP Layer**: Regex keyword matching and spatial geometry calculate target poses and clearance flanks with zero neural weights.
- **Learned Neural Layer**: 4-layer CNN feature extraction, latent transition foresight (Δs), and collision probability prediction.
- **Discovery Implication**: Mechanistic discovery must evaluate latent foresight directly without confounding it with engineered waypoints.



Operational Graph: Locating the FPGA Acceleration Boundary

One Core Insight

The 0.48 μs FPGA measurement isolates the MLP predictor core; end-to-end inference includes observation capture and CNN encoding.

- **PL Accelerated Core**: 3-layer MLP transition/policy network (6,788 - 8,609 MACs) running at 48 clock cycles (0.48 μs @ 100 MHz).
- **PS Host Tasks**: Camera rendering (2.1 ms) and CNN visual encoding (4.8 ms on ARM / 0.06 ms on GPU).
- **Circuit Extraction Target**: Phase 7 will investigate whether the visual encoder itself can be pruned into a compact circuit deployed to PL.

Operation	Hardware Domain	Latency
Camera Frame Capture	Host MuJoCo	$\approx$ 2.1 ms
Visual CNN Encoder	Host CPU / GPU	4.8 ms / 60 μs
AXI-Lite PS-PL Write	AXI4-Lite Bus	0.72 μs
Predictor PL Kernel	**KR260 FPGA PL**	**0.48 μs**
AXI-Lite PL-PS Read	AXI4-Lite Bus	0.64 μs
Control Timestep	Host AMR Loop	100 ms (10 Hz)

Literature Gap Analysis: Aligning Implementation with Theory

One Core Insight

Architectural downscaling to 148k parameters was essential for on-chip FPGA synthesis while preserving core JEPA and SIGReg principles.

- **LeWM (arXiv:2603.19312)**: Original 15M-param ViT exceeds Kria XCK26 on-chip BRAM (1.2 MB). Compact Conv4-LN faithfully preserves SIGReg moment matching and non-generative dynamics.
- **H-JEPA (arXiv:2610.06805)**: 2-tier implementation ($K = 5$ macro stride) matches physical JetAuto Mecanum dynamics (0.30 m clearance radius per macro step).
- **Methodology Verified**: Documented transparently in `research/audit/implementation_deviations.md`.

Theoretical Alignment

LeWM SIGReg Objective:

$$\mathcal{L} = \frac{1}{B} \sum \| \hat{s}_{t+1} - \mathrm{sg}(s_{t+1}) \|^2 + \lambda_{\mathrm{reg}} \mathcal{L}_{\mathrm{SIGReg}}(s)$$

H-JEPA Hierarchy:

$$\text{Macro: } \hat{S}_{k+1} = T_{\mathrm{macro}}(S_k, A_k^{\mathrm{macro}})$$

$$\text{Micro: } a_t = \pi_{\mathrm{micro}}(s_t, \hat{S}_{k+1})$$

Phase 0 Acceptance Verification & Readiness for Phase 1

One Core Insight

All five Phase 0 acceptance criteria are satisfied; the foundation is established for controlled dataset generation and neural instrumentation.

- **Criterion 1**: Every trainable parameter accounted for → **PASS**
- **Criterion 2**: Inputs/outputs identified down to shapes → **PASS**
- **Criterion 3**: Inference pathways fully traced → **PASS**
- **Criterion 4**: Checkpoints bit-exact reproducible → **PASS**
- **Criterion 5**: Implementation deviations documented → **PASS**

Next Phase Prerequisites (Phase 1)

1. Generate controlled counterfactual dataset (colors, positions, orientations).
2. Formalize 6 capability categories.
3. Isolate clean vs. corrupted pairs for causal activation patching.